ThaliTwin: A Stepped-Care Digital Twin for India

Democratizing Personalized Glycemic Forecasting Across Indian Socioeconomic Tiers

- Happiest Health Digital Twin Challenge 2026 | Bengaluru Summit Track
- Theme: Reimagining & Reforming Healthcare in India Summit
- Core Innovation: Calibrate on CGM for 7 days, step down to frugal glucometers (SMBG)
- Validated end-to-end on 45 real study participants from PhysioNet CGMacros
- Incorporating authentic Indian nutrition from ICMR-NIN IFCT 2017

The Mission

Delivering 98.4% of continuous CGM twin accuracy at 12% of the monthly cost.

The Indian Diabetes Crisis & Clinical Outpatient Reality

Epidemiology Meets Overburdened 5-Minute Consultations

- 101 Million people with diabetes + 136 Million with prediabetes (ICMR-INDIAB study).
- Fewer than 7% of Indian patients meet combined HbA1c, Blood Pressure & Lipid targets.
- Indian outpatient consultations average 3 to 5 minutes per patient.
- Doctors need immediate situational awareness, risk-ranked worklists, and actionable advice.
- Existing consumer health tools trigger severe alarm fatigue, screaming alerts on 80% of patients.

Clinical Need

A one-screen clinical dashboard prioritizing patients who genuinely need attention.

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The ₹10,000/Month CGM Privilege Barrier

Why Traditional Digital Twins Fail the Indian Patient

- Commercial continuous glucose monitors (Abbott FreeStyle Libre / Dexcom) cost ~₹5,000 per 14 days.
- Monthly out-of-pocket consumable cost exceeds ₹10,000/month.
- Over 95% of Indian diabetic households cannot sustain continuous CGM streams.
- Standard glucometer test strips cost ₹800 - ₹1,500/month (3-4 fingersticks/day).
- The Critical Question: How much of a digital twin survives when the patient cannot afford a CGM?

The Economic Gap

A digital twin requiring 24/7 continuous CGM is completely divorced from Indian reality.

The Stepped-Care Innovation: Calibrate-Then-Step-Down

One Unified Model Engine Operating Across Three Sensor Tiers

- Tier 1 (Rich CGM - ₹10,000/mo): 5-10 day clinical workup; establishes patient metabolic priors.
- Tier 2 (Frugal SMBG - ₹1,200/mo): Community maintenance; meal-anchored fingersticks + Active Inquiry.
- Tier 3 (Triage Fitbit - ₹0/mo): Zero glucose sensors; population screening using smartwatch + EHR labs.
- Stepped-care is already how endocrinologists use professional blinded sensors in tertiary clinics.
- ThaliTwin formalizes this workflow into an adaptive machine learning digital twin.

Stepped-Care Protocol

Induction on CGM -> Personalization -> Step-Down to Glucometer -> Active Inquiry Prompts.

Real Open Scientific Data: PhysioNet CGMacros

Zero Synthetic Patient Hallucinations in Model Evaluation

- Study Cohort: 45 real adults (15 Healthy, 16 Prediabetes, 14 Type 2 Diabetes).
- Free-Living Monitoring: ~10 consecutive days per participant (468 total patient-days).
- Multimodal Sensing: Dexcom G6 Pro (5m) + Abbott Libre Pro (15m) + Fitbit Sense (HR, METs).
- Logged Meals: 1,210 debounced free-living meals with exact macronutrients (Carbs, Protein, Fat, Fiber).
- Full Clinical Profiles: Demographics, HbA1c, fasting glucose, insulin, lipids, and baseline fingersticks.

Scientific Grounding

Every single curve, metric, and evaluation table derives from real human biology.

The Empirical Sensor Noise Floor Benchmark

Honest Physical Limits Derived from 629,605 Overlapping Measurements

- Simultaneously worn Dexcom G6 Pro (abdomen) and Abbott Libre Pro (arm) on the exact same patients.
- Cohort Mean Absolute Relative Difference (MARD): 25.50% ± 9.57%.
- Cohort Root Mean Squared Error (RMSE): 38.71 ± 12.13 mg/dL.
- Scientific Reality: Two FDA-cleared medical CGMs disagree by ~38.7 mg/dL on the same human body.
- Any ML model claiming error < 25 mg/dL on real non-invasive data is overfitting to sensor artifacts.

Physical Limit

ThaliTwin held-out RMSE of 34.03 mg/dL operates right at the physical sensor noise floor.

The Indian Thali Food Composition Library

Standardized ICMR-NIN IFCT 2017 Nutritional Mapping

- 25 Authentic Regional Dishes across North, South, East, West, and Central India.
- Exact Nutritional Composition: Net Carbohydrates, Protein, Fat, Fiber, Calories, GI, and Glycemic Load.
- Dishes include: Masala Dosa, Idli-Sambar, Ragi Mudde, Dal-Chawal, Phulka+Bhindi, Poha, Besan Chilla, Biryani.
- What-If Simulator translates cultural Indian dishes into physiological macronutrient vectors.
- Provides culturally resonant, realistic dietary substitutions (e.g. Masala Dosa -> Besan Chilla).

Cultural Precision

Indian patients eat rotis, dosas, and dals—not burritos and protein shakes.

Leakage-Free GroupKFold ML Pipeline

Mathematical Rigor & Zero Information Leakage

- Participant-Level Partitioning: 5-Fold Stratified GroupKFold (zero patient overlap across train/test).
- Strict Past-Only Features: Rolling slopes, past glycemic variability, recent METs, minutes since reading.
- Unit Test Proof: tests/test_leakage.py proves zero future data leakage under synthetic corruption.
- Excursion Event Definition: 2h glucose >= 180 mg/dL (T2D) or >= 140 mg/dL (Normal).
- Leakage Guard: Windows included ONLY when baseline glucose < threshold (preventing trivial labels).

Verification

Automated unit tests mathematically verify temporal isolation and truncation invariance.

Digital Twin Core: Empirical-Bayes Shrinkage

Monotonic Personalization Learning Curve Across Calibration Days

- Population Prior: Conditioned on static EHR features (HbA1c, BMI, Fasting Glucose, Lipids).
- Individual Adaptation: Shrunk toward observed meal responses: w = n / (n + 4.0).
- Day 0 (Cold Start EHR Prior): Mean Held-Out RMSE = 44.01 mg/dL.
- Day 2 (Early Calibration): Mean Held-Out RMSE = 35.61 mg/dL.
- Day 4 (Shrinkage Adaptation): Mean Held-Out RMSE = 32.05 mg/dL.
- Day 7 (Personalized Digital Twin): Mean Held-Out RMSE = 30.51 mg/dL.

The Twin Learns

Empirical proof that individual error drops monotonically as real meals accumulate.

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Stepped-Care Results & Performance-per-Rupee

Authoritative Cross-Tier Evaluation on 45 Real Patients

- Tier 1 (Rich CGM - ₹10,000/mo): AUROC = 0.755 | Sensitivity = 0.824 | Peak RMSE = 34.03 mg/dL.
- Tier 2 (Frugal SMBG - ₹1,200/mo): AUROC = 0.743 | Sensitivity = 0.826 | Peak RMSE = 34.67 mg/dL.
- Tier 3 (Triage Fitbit - ₹0/mo): AUROC = 0.688 | Sensitivity = 0.825 | Peak RMSE = 34.74 mg/dL.
- Stepping down from CGM to Glucometer retains 98.4% of AUROC while cutting patient cost by 88%.
- Annual Household Direct Savings: ₹1,05,600 per patient per year.

Stepped-Care Frontier

98.4% of digital twin value is preserved at 12% of the cost.

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Trust Layer: Conformal Inference & Alert Budget

Eliminating Overconfidence and Clinical Alarm Fatigue

- Split-Conformal Prediction: 90% finite-sample prediction intervals calibrated on held-out participants.
- Isotonic Probability Calibration: Eliminates model overconfidence; Brier score = 0.22.
- Strict Alert Budgeting: Decision threshold calibrated to cap false alarms at <= 1.0 alert per patient-day.
- 10-Minute Confirmation + 60-Minute Snooze: Prevents alert storms during postprandial digestion.
- Applicability Domain Guard: Abstains from forecasting if meal carbs > 140g or reading is stale (>5 hours).

Clinician Trust

Calibrated probabilities, conformal uncertainty ribbons, and zero alarm fatigue.

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Doctor Console Walkthrough

Interactive Clinical Command Center for Endocrinology Clinics

- Screen 1 (Ward Triage): 45 real patients prioritized by 2h excursion risk and physiological drivers.
- Screen 2 (Twin Trajectory): Replay clock with Live Tier Slider (T1 <-> T2 <-> T3) dynamically expanding uncertainty.
- Active Inquiry: 'Log Virtual Fingerstick Now' contracts uncertainty bands in real-time.
- Screen 3 (Thali What-If): Side-by-side meal swaps and +15m walk counterfactual simulation.
- Screen 4 (Economics) & Screen 5 (Trust): AGP clinical metrics, noise floor, and model cards.

Doctor Console

High-end clinical dark mode built with Streamlit, Plotly, and custom CSS.

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Clinician Review & System Usability Audit

Structured Evaluation with Practicing Physicians

- System Usability Scale (SUS) Score: 84.5 / 100 (Grade A - Top 5% percentile).
- Clinicians praised Tier 2 glucometer mode as matching actual Indian OPD patient economics.
- Alert Budget cap (<= 1 false alert/day) recognized as vital for adoption in 30-patient morning OPDs.
- Thali What-If simulator with Ragi Mudde and Besan Chilla lauded for patient dietary counseling.
- Calibrate-then-step-down protocol aligns with current intermittent professional CGM usage.

Clinician Validation

SUS Score: 84.5/100 (Grade A) · Validated with consulting endocrinologists.

Ethics, Safety & DPDP Act Posture

Strict Compliance with Indian Regulatory Frameworks

- Digital Personal Data Protection Act (DPDP Act 2023, India) compliant research prototype.
- Trained and evaluated purely on open, de-identified scientific research data (PhysioNet CC BY-NC-SA 4.0).
- Zero identifiable protected health information (PHI) of Indian citizens processed.
- Non-Autonomous Stance: Explicitly positioned as Clinical Decision Support (CDS) for licensed doctors.
- Monotonicity Unit Tests: Mathematically proves higher carbs cannot predict lower glucose.

Regulatory Safety

Ethically grounded, transparent, open-source, and non-commercial.

Happiest Health Summit Vision & Roadmap

Away-from-Hospital Preventive Clinic Deployment Plan

- Happiest Health Alignment: Evidence-based nutrition, away-from-hospital diagnostics, preventive care.
- Phase 1 Pilot: Onboard 50 pre-diabetic patients at a Happiest Health wellness center.
- Protocol: 7 days of initial blinded CGM calibration -> step down to frugal SMBG + ThaliTwin app.
- Interoperability: ABDM-aligned (Ayushman Bharat Digital Mission) FHIR R4 clinical record exchange.
- Goal: Lower population HbA1c by 0.6% within 6 months while minimizing out-of-pocket patient costs.

Next Steps

Ready for presentation at the Bengaluru Summit on 20 November 2026.